Course: DATA602 - Principles of Data Science
Assignment: Paper Writing 02
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2. Paper Writing Assignment (Points 20):

Read the Paper *"Randomized Placebo-Controlled Placebo Trial to Determine the Placebo Effect Size"* and address the following questions:

Summarize the Key Points: What are the main objectives of the paper?

The paper talks about how the patients' belief that they are getting effective treatment affects their pain relief. Patients with chronic plantar heel pain divided in 2 groups receiving similar placebo treatment were studied by researchers, the only difference was that one group was told that the treatment was real and effective while the other was told that it was a placebo. This helped the researches to examine effect of treatment expectations

The authors also aimed to demonstrate the Inverse Placebo Randomized Controlled Trial (IPRCT) design for estimating placebo effect. It was believed that patients who believed were receiving real treatment would get greater pain relief.

What is the significance of using a placebo-controlled trial design in this study, and how does it contribute to understanding the placebo effect?

The placebo controlled design helps in understanding the influence of patients belief and expectations on the pain relief. Both groups received the same placebo treatment but one group was told that their pain was real and effective while the other was told it was a placebo. Keeping the procedure same but randomly assigning patients helped in controlling questions that may arise from difference in outcomes.

Patients who believed that they received real treatment reported greater relief in pain after 6 weeks. This supports the idea that expectations and belief in the treatment can have effect and have influence on the perceived pain. This placebo experiment helps us understand and distinguish the effect of treatment expectations from changes that may occur naturally overtime.

Write a summary of the control, treatment, and sample of study from this paper.

Controlled: The placebo-placebo group consisted of 53 patients who received false ESWT (extracorporeal shockwave treatment). They were informed that the treatment was false and would have no effect.

Treatment: The Placebo-verum group consisted of 53 patients who received the same false treatment of ESWT (extracorporeal shockwave treatment) but they were told it was real and effective. Thus the experimental difference was the information given to the patients, neither group received the ESWT.

Sample of study: 182 patients were assessed out of which 106 were enrolled, including 64 women aged 31-75 years with average 50.1 years. Patients had chronic plantar heel pain and atleast 6 months or unsuccessful conservatory treatment. They were randomly assigned to 2 groups and had their pain measured at baseline and in 6 weeks after treatment using VAS (Visual Analog Scale) and RMS (Roles and Maudsley Scale).

What are the study's key findings, and how do they contribute to the current understanding of placebo effects in clinical trials?

The study found that the patients who thought they were receiving real and effective treatment experienced greater reported pain relief than the patients who knew it was a placebo even though both the groups received the same false treatment. After 6 weeks the average improvement in Visual Analog Scale scores was of 1.5 points versus 0.38 points while improvement in Roles and Maudsley Scale scores was of 0.5 points versus 0.1 points. The difference in improvement was statistically significant for both measures.

These findings support the role of expectations and belief in placebo-related pain relief. By comparing identical procedures with different treatment information this study helps in distinguish expectation related effects from natural improvement.

As a Data Scientist, if you were tasked with designing an experiment for the placebo-controlled study described in this paper, would you propose any alternative design or approach? Please provide your reasoning for any proposed changes

As a Data Scientist I would propose 4 randomized groups combining the active and placebo treatment with being told that the treatment was active or placebo. This would help in separating the actual treatment effect from the effects of expectations and see if they interact and have other effects and also extend the follow up to see if the pain relief lasts.